Assessment 7

1 D, 2 D and N D Analysis

1. Load the dataset shared via email

2. Compute summary statistics: Mean, Median, Minimum, Maximum, Range, Variance, Standard Deviation, Interquartile Range (IQR), Skewness (using e1071::skewness), and Kurtosis (using e1071::kurtosis).

3. Store these values in a single structured data.frame or named vector and print it.

4. Create a frequency distribution table and a relative frequency distribution table for a categorical variable. Generate a Pie Chart and a Bar Plot to represent the distribution.

5. Create a frequency distribution table with bin intervals for a numerical variable. Plot its Histogram and a Box Plot to check for outliers.

6. Choose two categorical variables: Construct a Contingency Table (Cross-tabulation) and compute the row/column percentages using prop.table()

7. Create a Stacked Bar Plot or Grouped Bar Plot to visualize the relationship.

8. Select one categorical variable and one numerical variable: Compute the mean, median, and standard deviation of the numerical variable grouped by each category using aggregate()

9. Generate Side-by-Side Box Plots and a Violin Plot to visualize the numerical distribution across categories.

10. Select two numerical variables: Calculate the Pearson and Spearman Correlation Coefficients (cor()) and Construct a Scatter Plot with a fitted linear regression trend line.

11. Select two categorical features and one numerical feature: Calculate summary statistics (Mean and Standard Deviation) for the numerical variable across all combinations of the two categorical variables and Create a Faceted Box Plot and a Grouped Bar Chart with Error Bars.

12. Select one categorical variable and two numerical variables: Construct a Scatter Plot of the two numerical features where points are colored/shaped by the categorical feature. compute a correlation matrix between the two numerical variables separately for each category level.

13. Select at least 3 numerical features from the dataset: Generate a full Correlation Matrix and plot it as a Heatmap

14. Produce a Pairs Plot / Scatterplot Matrix to examine pairwise relationships and distributions simultaneously.